

Probability & Statistics (MS-301) Software Engineering

Lecture # 14-15 (Random Variables and Probability Distributions)

Basic Definitions

A **random variable** is a function that associated a real number with each element in the sample space.

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **Discrete Sample**.

If a sample space contains a infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

Discrete Distribution

The set of ordered pair $(x, f(x))$ is a probability function , Probability mass function or probability distribution of the **discrete** random variable X if, for each possible outcome x

1. $f(x) \geq 0,$
2. $\sum_x f(x) = 1,$
3. $P(X = x) = f(x).$

The **Cumulative frequency distribution** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

Continuous Distribution

The set of ordered pair $(x, f(x))$ is a probability function , Probability mass function or probability distribution for the continuous random variable X , define over the set of real numbers R , if

1. $f(x) \geq 0$, for all $x \in R$.

2. $\int_{-\infty}^{\infty} f(x) dx = 1$.

3. $P(a < X < b) = \int_a^b f(x) dx$.

The **Cumulative frequency distribution** $F(x)$ of a continuous random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad \text{for } -\infty < x < \infty.$$

3.5 Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X :

(a) $f(x) = c(x^2 + 4)$, for $x = 0, 1, 2, 3$;

(b) $f(x) = c \binom{2}{x} \binom{3}{3-x}$, for $x = 0, 1, 2$.

Solution 3.5

(a)

$$f(x) = c(x^2 + 4) : \text{ for } x = 0, 1, 2, 3$$

$$f(0) = c\{0^2 + 4\} = 4c$$

$$f(1) = c\{1^2 + 4\} = 5c$$

$$f(2) = c\{2^2 + 4\} = 8c$$

$$f(3) = c\{3^2 + 4\} = 13c$$

$$f(0) + f(1) + f(2) + f(3) = 1$$

$$4c + 5c + 8c + 13c = 1$$

$$30c = 1 \Rightarrow \boxed{c = \frac{1}{30}}$$

$$(b) \quad f(x) = e \binom{2}{x} \binom{3}{3-x} \text{ for } x = 0, 1, 2$$

$$f(0) = e \binom{2}{0} \binom{3}{3-0}$$

$$= e \binom{2}{0} \binom{3}{3}$$

$$= e (1)(1)$$

$$= e$$

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

$$f(1) = e \binom{2}{1} \binom{3}{3-1}$$

$$= e \binom{2}{1} \binom{3}{2}$$

$$= e \frac{2!}{1!(2-1)!} \times \frac{3!}{2!(3-2)!}$$

$$e (2)(3)$$

$$f(1) = 6c$$

$$f(2) = c \binom{2}{2} \binom{3}{3-2}$$

$$= c(1)(3)$$

$$= 3c$$

$$f(n) = f(0) + f(1) + f(2) = 1$$

$$c + 6c + 3c = 1$$

$$10c = 1$$

$$\boxed{c = \frac{1}{10}}$$

3.6 The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.

Q. solution 3.6

① at least 200 days:

$$\begin{aligned} P(X \geq 200) &= 20000 \int_{200}^{\infty} \frac{1}{(x+100)^3} dx \\ &= 20000 \int_{200}^{\infty} (x+100)^{-3} dx \\ &= 20000 \left| \frac{(x+100)^{-3+1}}{-3+1} \right|_{200}^{\infty} \\ &= \cancel{20000} \left| \frac{(x+100)^{-2}}{\cancel{-2}} \right|_{200}^{\infty} \\ &= -10000 \left| \frac{1}{(x+100)^2} \right|_{200}^{\infty} \\ &= -10000 \left\{ \frac{1}{(\infty+100)^2} - \frac{1}{(200+100)^2} \right\} \\ &= -10000 \left\{ 0 - \frac{1}{90000} \right\} \\ &= \frac{10000}{90000} = \frac{1}{9} \end{aligned}$$

⑤ 3.6
(b) anywhere from 80 to 120 days

$$P(80 < x < 120) = 20000 \int_{80}^{120} \frac{1}{(x+100)^3} dx$$

$$= 20000 \left| \frac{(x+100)^{-2}}{-2} \right|_{80}^{120}$$

$$= -10000 \left| \frac{1}{(x+100)^2} \right|_{80}^{120}$$

$$= -10000 \left| \frac{1}{(120+100)^2} - \frac{1}{(80+100)^2} \right|$$

$$= -10000 \left| \frac{1}{48400} - \frac{1}{32400} \right| = \frac{1000}{9801}$$

3.7 The total number of hours, measured in units of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner

- (a) less than 120 hours;
- (b) between 50 and 100 hours.

37 solution

a) less than 120 hours.

$$P(X < 1.2) = \int_0^1 x + \int_1^{1.2} (2-x) dx$$

$$= \left| \frac{x^2}{2} \right|_0^1 + \left| 2x - \frac{x^2}{2} \right|_1^{1.2}$$

$$= \left\{ \frac{(1)^2}{2} - \frac{(0)^2}{2} \right\} + \left\{ \left(2(1.2) - \frac{(1.2)^2}{2} \right) - \left(2(1) - \frac{(1)^2}{2} \right) \right\}$$

$$= \frac{1}{2} + \left[2.4 - 0.72 - 2 + \frac{1}{2} \right]$$

$$= 0.5 + 0.18$$

$$= 0.68$$

⑤ between 50 and 100 hours

$$P(0.5 < x < 1) = \int_{0.5}^1 x \, dx$$

$$= \left| \frac{x^2}{2} \right|_{0.5}^1$$

$$= \frac{1}{2} - \frac{0.25}{2}$$

$$= \frac{3}{8}$$

3.11 A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. If x is the number of defective sets purchased by the hotel, find the probability distribution of X . Express the results graphically as a probability histogram.

Solution 3.11

$$n = 7$$

$$\text{defective} = 02$$

$$\text{perfect} = 05$$

$$\text{selected} = 03$$

$$f(x) = \frac{\binom{2}{x} \binom{5}{3-x}}{\binom{7}{3}} \quad ; x = 0, 1, 2, 3$$

$$f(0) = \frac{\binom{2}{0} \binom{5}{3-0}}{\binom{7}{3}} = \frac{\frac{2!}{0!(2-0)!} \times \frac{5!}{3!(5-3)!}}{\frac{7!}{3!(7-3)!}}$$

$$f(0) = \frac{(1)(10)}{35} = \frac{10}{35} = \frac{2}{7}$$

$$f(1) = \frac{\binom{2}{1} \binom{5}{2}}{\binom{7}{3}} = \frac{(2)(10)}{(35)} = \frac{4}{7}$$

$$f(2) = \frac{\binom{2}{2} \binom{5}{1}}{\binom{7}{3}} = \frac{(1)(5)}{35} = \frac{1}{7}$$

x	0	1	2
f(x)	2/7	4/7	1/7

3.12 An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7, \end{cases}$$

find

- (a) $P(T = 5)$;
- (b) $P(T > 3)$;
- (c) $P(1.4 < T < 6)$;

solution
3.12

$$\begin{aligned}\textcircled{a} \quad P(T=5) &= F(5) - F(3) \\ &= \frac{1}{2} - \frac{1}{4} \\ &= \frac{1}{4}\end{aligned}$$

$$\begin{aligned}\textcircled{b} \quad P(T > 3) &= 1 - P(T \leq 3) \\ &= 1 - F(3) \\ &= 1 - \frac{1}{4} = \frac{3}{4}\end{aligned}$$

$$\begin{aligned}\textcircled{c} \quad P(1.4 < T < 6) &= F(6) - F(1.4) \\ &= \frac{3}{4} - \frac{1}{4} \\ &= \frac{2}{4} \\ &= \frac{1}{2}\end{aligned}$$

3.13 The probability distribution of X , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct the cumulative distribution function of X .

sol 3.13

$$f(0) = f(1) = 0.41$$

$$\begin{aligned} F(1) &= f(0) + f(1) \\ &= 0.41 + 0.37 \\ &= 0.78 \end{aligned}$$

$$\begin{aligned} F(2) &= f(0) + f(1) + f(2) \\ &= 0.41 + 0.37 + 0.16 \end{aligned}$$

$$F(2) = 0.94$$

$$\begin{aligned} F(3) &= f(0) + f(1) + f(2) + f(3) \\ &= 0.99 \end{aligned}$$

$$\begin{aligned} F(4) &= f(0) + f(1) + f(2) + f(3) + f(4) \\ &= 1 \end{aligned}$$

function

$$F(x) = \begin{array}{ll} 0 & x < 0 \\ 0.41 & 0 \leq x \leq 1 \\ 0.78 & 1 \leq x < 2 \\ 0.94 & 2 \leq x < 3 \\ 0.99 & 3 \leq x < 4 \\ 1 & x \geq 4 \end{array}$$

3.14 The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X ;
- (b) using the probability density function of X .

Solution 3.14

$$1 \text{ Hour} = 60 \text{ mins}$$

$$1 \text{ minute} = \frac{1}{60} \text{ Hours}$$

$$12 \text{ minutes} = \frac{1 \times 12}{60} \text{ Hours}$$

$$12 \text{ minutes} = 0.2 \text{ Hours}$$

(i), using the $F(x)$ of x

$$P(X < 0.2) = 1 - e^{-8(0.2)}$$

$$= 1 - e^{-1.6}$$

$$= 0.7981$$

ii) using the Probability density function
of X

$$P(X < 0.2) = ?$$

Convert $F(u)$ to $f(u)$

$$f(u) = \frac{d}{du} (F(u))$$

$$f(u) = \frac{d}{du} \{1 - e^{-8u}\}$$

$$= -(-8)e^{-8u}$$

$$f(u) = 8e^{-8u} \quad u \geq 0$$

0 elsewhere

$$P(u < 0.2) = 8 \int_0^{0.2} e^{-8u} du$$

$$= 8 \left| \frac{e^{-8u}}{-8} \right|_0^{0.2}$$

$$= - \left[e^{-8(0.2)} - e^{-8(0)} \right]$$

$$= 1 - e^{-8(0.2)}$$

$$= 0.7981$$

3.15 Find the cumulative distribution function of the random variable X representing the number of defectives in Exercise 3.11. Then using $F(x)$, find

- (a) $P(X = 1)$;
- (b) $P(0 < X \leq 2)$.

Solution 3.15

x	0	1	2
$f(x)$	$\frac{2}{7}$	$\frac{4}{7}$	$\frac{1}{7}$

$$F(0) = f(0) = \frac{2}{7}$$

$$F(1) = f(0) + f(1)$$

$$= \frac{4}{7} + \frac{2}{7} = \frac{6}{7}$$

$$F(2) = f(0) + f(1) + f(2)$$

$$= \frac{4}{7} + \frac{2}{7} + \frac{1}{7} = \frac{7}{7} = 1$$

$F(x)$

$$0 \quad x < 0$$

$$\frac{2}{7} \quad 0 \leq x < 1$$

$$\frac{6}{7} \quad 1 \leq x < 2$$

$$1 \quad x \geq 2$$

$$\begin{aligned} \textcircled{a} \quad P(X=1) &= F(1) - F(0) \\ &= \frac{6}{7} - \frac{2}{7} = \frac{4}{7} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad P(0 < X \leq 2) &= F(2) - F(0) \\ &= 1 - \frac{2}{7} = \frac{5}{7} \end{aligned}$$

3.17 A continuous random variable X that can assume values between $x = 1$ and $x = 3$ has a density function given by $f(x) = 1/2$.

- (a) Show that the area under the curve is equal to 1.
- (b) Find $P(2 < X < 2.5)$.
- (c) Find $P(X \leq 1.6)$.

solution 3.17

$$f(u) = \begin{cases} \frac{1}{2} & 1 \leq u \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

① Show that the area under the curve is equal

$$\begin{aligned} f(u) &= \frac{1}{2} \int_1^3 du \\ &= \frac{1}{2} \left| u \right|_1^3 \\ &= \frac{1}{2} |3 - 1| \\ &= \frac{1}{2} (2) \\ &= 1 \end{aligned}$$

$$\begin{aligned}
 \textcircled{b} \quad P(2 < X < 2.5) &= \frac{1}{2} \int_2^{2.5} du \\
 &= \frac{1}{2} \left| u \right|_2^{2.5} \\
 &= \frac{1}{2} |2.5 - 2| \\
 &= \frac{1}{2} (0.5) = \frac{1}{2} \left(\frac{1}{2} \right) \\
 &= \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{c} \quad P(X \leq 1.6) &= \frac{1}{2} \int_1^{1.6} du \\
 &= \frac{1}{2} \left| u \right|_1^{1.6} \\
 &= \frac{1}{2} |1.6 - 1| \\
 &= \frac{3}{10}
 \end{aligned}$$

3.21 Consider the density function

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Evaluate k .
- (b) Find $F(x)$ and use it to evaluate

$$P(0.3 < X < 0.6).$$

solution 3.21

Ⓐ Evaluate k .

$$k \int_0^1 x^{1/2} dx = 1$$

$$k \left| \frac{x^{1/2+1}}{\frac{1}{2}+1} \right|_0^1 = 1$$

$$k \left| \frac{x^{3/2}}{3/2} \right|_0^1 = 1$$

$$\frac{2k}{3} \left| x^{3/2} \right|_0^1 = 1$$

$$\frac{2k}{3} \left| (1)^{3/2} - (0)^{3/2} \right| = 1$$

$$\frac{2k}{3} (1) = 1$$

$$\boxed{k = \frac{3}{2}}$$

$$f(x) = \frac{2}{3} \sqrt{x} \quad 0 < x < 1$$

0 elsewhere

$$F(x) = \frac{2}{3} \int_0^x (u)^{1/2} du$$

$$= \frac{2}{3} \left| \frac{x^{3/2}}{3/2} \right|_0^x$$

$$= \frac{2}{3} \times \frac{2}{3} \left| x^{3/2} - 0 \right|$$

$$f(x) = x^{3/2} \quad 0 < x < 1$$

0 elsewhere

$$\begin{aligned} P(0.3 < x < 0.6) &= F(0.6) - F(0.3) \\ &= (0.6)^{3/2} - (0.3)^{3/2} \\ &= 0.3004 \end{aligned}$$

3.24 Find the probability distribution for the number of jazz CDs when 4 CDs are selected at random from a collection consisting of 5 jazz CDs, 2 classical CDs, and 3 rock CDs. Express your results by means of a formula.

3.24 solution

Probability distribution for the number of Jazz CDs.

Selected # 04 CDs

Jazz → 05 CDs

Classic → 02 CDs

Rock → 03 CDs

10 CDs

$$f(x) = \frac{\binom{5}{x} \binom{5}{4-x}}{\binom{10}{5}}; \quad x = 0, 1, 2, 3, 4$$

3.29 An important factor in solid missile fuel is the particle size distribution. Significant problems occur if the particle sizes are too large. From production data in the past, it has been determined that the particle size (in micrometers) distribution is characterized by

$$f(x) = \begin{cases} 3x^{-4}, & x > 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that this is a valid density function.
- (b) Evaluate $F(x)$.
- (c) What is the probability that a random particle from the manufactured fuel exceeds 4 micrometers?

Solution 3.29

① Verify that this is a valid density function

$$3 \int_1^{\infty} x^{-4} dx = 1$$

$$\Rightarrow 3 \left| \frac{x^{-4+1}}{-4+1} \right|_1^{\infty} = 1$$

$$\Rightarrow 3 \left| \frac{x^{-3}}{-3} \right|_1^{\infty} = 1$$

$$= - \left| \frac{1}{x^3} \right|_1^{\infty} = 1$$

$$= - \left| \frac{1}{(\infty)^3} - \frac{1}{1} \right|$$

$$= - | 0 - 1 |$$

$$= 1$$

② Evaluate $F(x)$

$$F(x) = 3 \int_1^x x^{-4} dx$$

$$= 3 \left| \frac{x^{-3}}{-3} \right|_1^x$$

$$= - \left| x^{-3} \right|_1^x$$

$$= - \left| x^{-3} - 1 \right|$$

$$F(x) = \begin{cases} 1 - x^{-3} & x > 1 \\ 0 & \text{elsewhere} \end{cases}$$

for $F(x)$ $= 1 - P(X < 4)$

$$\begin{aligned}\textcircled{c} P(X > 4) &= 1 - F(4) \\ &= 1 - \{1 - (4)^{-3}\} \\ &= 1 - 1 + \frac{1}{4^3} \\ &= 0.015625\end{aligned}$$

for $f(x)$

$$\begin{aligned}\textcircled{c} P(X > 4) &= 3 \int_4^{\infty} x^{-4} dx \\ &= 3 \left| \frac{x^{-3}}{-3} \right|_4^{\infty} \\ &= - \left| \frac{1}{x^3} \right|_4^{\infty} \\ &= - \left| 0 - \frac{1}{4^3} \right| \\ &= \frac{1}{4^3} = 0.015625\end{aligned}$$

Questions?

